# Invisible Bits Consulting FZCO — Capability Statement

Invisible threats, made visible — zero-knowledge & privacy engineering, compliant asset tokenization and post-quantum readiness, with protocol security reviews taken on selectively.

Invisible Bits Consulting FZCO is a Dubai-based blockchain security and applied-cryptography consultancy. We lead with consulting engineering — zero-knowledge and privacy engineering, compliant real-world asset tokenization, and post-quantum readiness — and take on protocol security reviews selectively, where the mechanism is novel enough to need one. Behind the firm stands a record of 40+ security reviews across Ethereum, Solana and Layer-2 ecosystems, a production zk-rollup shipped to mainnet, and a working national-scale tokenization proof of concept for the Saudi Real Estate Registry.

## TRACK RECORD AT A GLANCE

| 40+ | 200+ | $50B+ | 15 |
|---|---|---|---|
| Security reviews led | Vulnerabilities found | Value secured at review | Published research works (dblp) |

## SERVICES

**Zero-Knowledge Engineering.** Circuit design, soundness review and prover engineering across Halo2, SP1, RISC0, ZISK and PICO — delivered as code your team runs and owns.

**Post-Quantum Readiness.** Cryptographic bill of materials across your estate, exposure ranked by data lifetime, and a costed migration plan to NIST FIPS 203, 204 and 205 — written for the board and the regulator.

**Architecture Review.** An independent, vendor-neutral read on where a blockchain platform genuinely earns its place, where a database is the honest answer, and a staged roadmap either way. We resell nothing.

**Real-World Asset Tokenization.** ERC-3643 architecture where the regulator's rules execute inside the token — eligibility, ownership caps and lock-ups checked before any transfer settles. Built for GCC regulatory regimes.

**Privacy Engineering.** Zero-knowledge proofs, secure multi-party computation and selective disclosure: counterparties verify eligibility or solvency without seeing the records behind it. Proven on an MPC health-data platform.

**Protocol Security Review (selective).** Deep reviews for novel mechanisms: a threat model for your design, severity-ranked findings with concrete exploit paths, and a re-test once fixes land.

## FLAGSHIP ENGAGEMENTS

**Zircuit zk-rollup — ZK lead**
Led five Rust engineers and three cryptographers through three generations of SNARK provers and multi-zkVM support, delivering 10× prover gains and mainnet in 14 months — $600M max TVL, zero security incidents during tenure; Trail of Bits and Quantstamp reviews cleared with no critical findings.

**Saudi Real Estate Registry (RER)**
National tokenization architecture with a working ERC-3643 proof of concept: compliant fractional property markets with a regulated exchange and stablecoin settlement — the registry, not the market, as policy owner. Architecture whitepaper, Aug 2026.

**Solana core protocol ×3**
Vulnerability research on the core runtime and consensus layers across three separate core-protocol audits — $15B+ TVL at time of audit.

## WIDER AUDIT & DELIVERY PORTFOLIO

Also audited: Lido stETH, Ondo Finance ×3, Avalanche, Badger, Tokemak, KeeperDAO, Reflexer, Mythical Games, SuperRare, Merit Circle, APWine and Kresko. Reports were issued under Quantstamp's name; six public reports name our principal: Ichi, Badger ibBTC, Tokemak, Reflexer, SuperRare and PlaySwoops. Prior-role delivery also includes blockchain PKI shipped in a production secure messenger and an MPC platform for health-data collaboration.

Engagements before 2025 were led by our founder in prior roles at Quantstamp, Zircuit and SAP; Invisible Bits was founded in 2025. TVL figures are as of the time of each engagement. An audit is a point-in-time review, not an endorsement.

## LEADERSHIP

**Founder, Director & Principal Security Researcher** — Ph.D. in Computer Science, TU Munich (summa cum laude). Former Head of Cloud Security at SAP, senior research engineer at Quantstamp and lead ZK cryptography engineer at Zircuit; built anti-tampering defenses for the Chromium browser in a research collaboration with Google (CODASPY 2017). Fifteen published research works on software security, compiler-based protection and zero-knowledge systems. Engagement teams are assembled per scope from a vetted network of cryptographers, protocol engineers and infrastructure architects.

## TALKS & PUBLICATIONS

Speaker at EthCC[8] (2025, "Optimizing ZK-Rollups: Unlocking Cost-Effective Proving Infrastructure") and EthCC[9] (2026, "Proving Less, Verifying More: 3.5× Faster L2 Validity Proofs"); two Ethproofs contributions; 15 research works (dblp) including ACSAC and CODASPY; architecture whitepaper on policy-driven real-estate tokenization for national land registries (2026). Full record: scholar.google.com/citations?user=8rWVCAIAAAAJ

## ENGAGEMENT MODEL & CONTACT

A scoping conversation — under NDA from the first call if you prefer — leads to a fixed-fee first engagement with one approvable number: most reviews run three to six weeks and close with a written, severity-ranked report and a re-test once fixes land. Larger builds are milestone-based and follow a completed review.

This statement can be tailored to specific RFP/RFI requirements, including DIFC, ADGM, DMCC, SAMA and CMA; technical annexes and references are available on request.


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